CML-103

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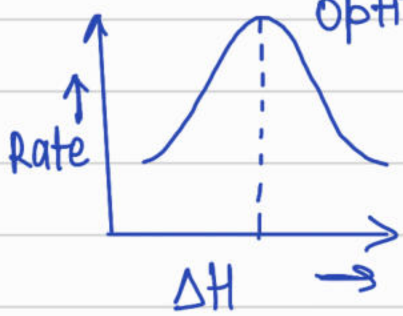
Book :- • Adamson & Gast → Physical chemistry of surface

• Principles of colloids & Heimenz & Rajagopal.

Topics :- Interface. i) Electrified solids
ii) Non Electrified Solid surface

- Adsorption isotherms, Helmohotz.
- Solid-gas, Solid-liquid interface.
- Sabatier principle.
- d-band theory.



$n = F(T, P)$

$n = F_T(P) \rightarrow$ isotherm

$n = F_P(T) \rightarrow$ isobar

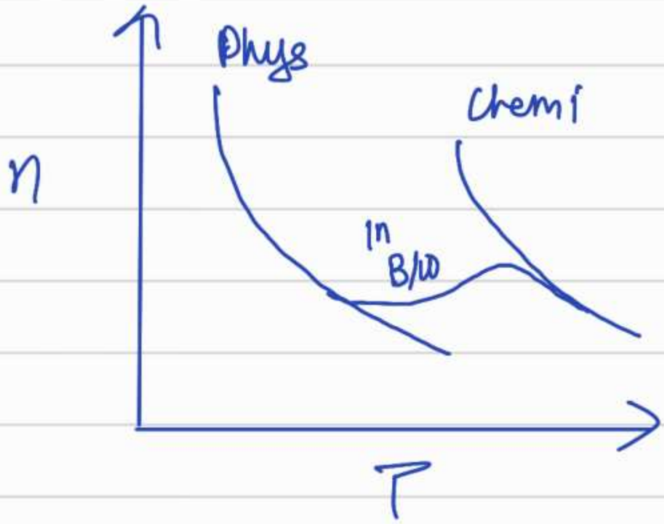
P vs T at fixed n ⇒ Isosterec

Physisorption	Chemisorption
→ Van der Wall	Chemical
→ 20 - 40 kJ mol^{-1}	200 - 400 kJ mol^{-1}
→ equilibrium fast	slow or fast may

→ equilibrium fast
reversible | slow or fast, may not be reversible.
→ Multilayer | monolayer.

<u>Solids</u> :- <u>Molecular</u> | <u>Refractory</u>

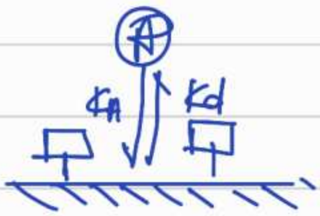
	Molecular	Refractory
	→ solid	solid
	→ ice, polymer	metals, metal oxides.
Physisorption →	Surface Restructuring	No surface resturcturing.
Chemisorption →	"	may or may not



Adsorption

Langmuir adsorption isotherm

$$\theta = \frac{bP}{1+bP} = \frac{KP}{1+KP}$$



(i) loving Langmuir

Total no. of sites $= S$
No. of sites occupied $= S_1$
Available sites $= S_0 = S - S_1$
θ = Coverage

Rate of ads $= k_a S_0 P_A = k_a P_A (S - S_1)$
Rate of des $= k_d S_1$

At equilibrium

$$k_a P_A S - k_a P_A S_1 = k_d S_1$$

$$\frac{S_1}{S} = \theta_A, \quad \frac{k_a}{k_d} = K = b$$

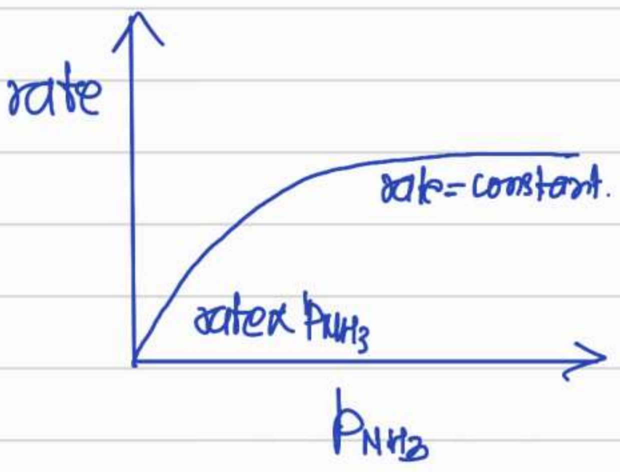
$$\Rightarrow \boxed{\theta = \frac{bP_A}{1 + bP_A}}$$

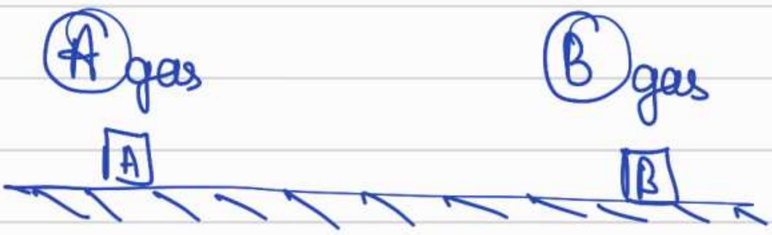
$\text{rate} = k_r \theta$ (If rxn happening at surface conc^n is imp & if it is happening in gas that press is imp.)

$$\text{rate} = k_r \theta = \frac{k_r b P_{NH_3}}{1 + bP_{NH_3}}$$

1) Very High P_{NH_3}
$\text{rate} = k_r$

2) Low $P_{NH_3} \Rightarrow \text{rate} = k_r b P_{NH_3}$





Total No. of site $= S$
site occupied by $A = S_A$

" " $B = S_B$

Tot. no. of unocc site $= S - S_A - S_B$

rate of ads of $A = k_a P_A (S - S_A - S_B)$

" " des of $B = k_d S_A$

$$k_d S_A = k_a P_A (S - S_A - S_B)$$

$$\frac{S_A}{S} = \theta_A, \quad \frac{S_B}{S} = \theta_B \text{ and } \frac{k_a^A}{k_d^A} = b_A,$$

$$\frac{k_a^B}{k_d^B} = b_B$$

$$\boxed{\theta_A = \frac{b_A P_A}{(1 + b_A P_A + b_B P_B)}}$$

$$\boxed{\theta_B = \frac{b_B P_B}{1 + b_A P_A + b_B P_B}}$$

* **Monolayer = All space occupied.**

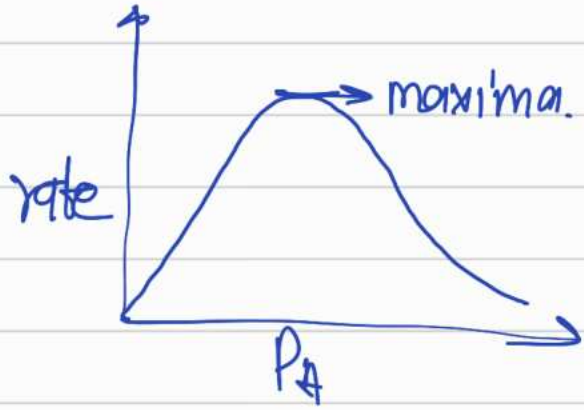
Langmuir - Hinshelwood mechanism.

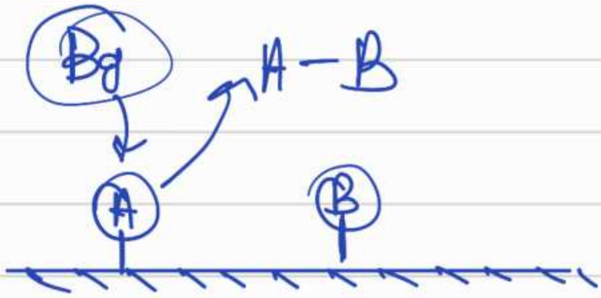
$$\text{rate} = k_r \theta_A \theta_B = \frac{k_r b_A P_A b_B P_B}{(1 + b_A P_A + b_B P_B)^2}$$

at low $P_A \Rightarrow$ rate $= k_r b_A P_A b_B P_B$

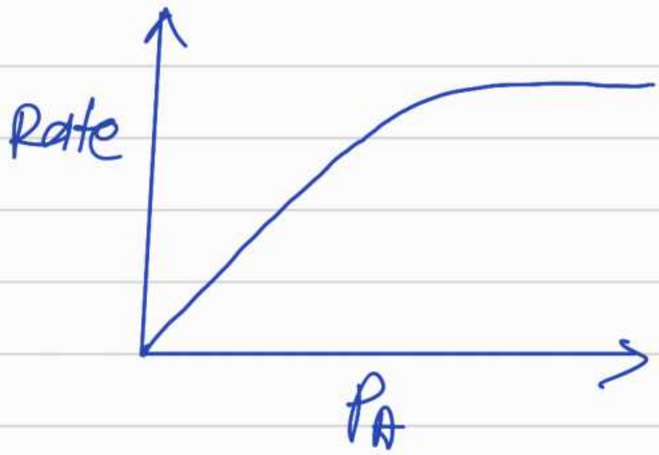
$(1+b_B P_B)^2$

at hight P_A, rate $\propto \frac{1}{P_A}$



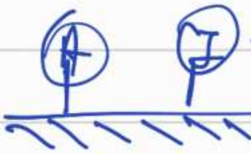


Langmuir-Rideal / Eley-Rideal.



rate $= k_r \theta_A P_B$

rate $= \dfrac{k_r P_B b_A}{(1+b_A P_A + P_B P_B)}$

Q -  → not reacting   find Rate

Biomolecular reaction b/w non-competing adsorbates

Total no. of sites for A $= S_1$ for B $= S_2$

sites occupied by A $= S_A$ $= S_B$

Available sites for A $= S_1 - S_A$ $= S_2 - S_B$

Rate of ads for A $= k_a P_A (1-\theta_A)\sigma_0 \rightarrow$ con^n/area

" " dis " " $= k_d \theta_A \sigma_0 \rightarrow$ con^n/area

$$Rate = k_r \theta_A \theta_B$$

$$= \frac{k_r b_A P_A \; b_B P_B}{(1+b_A P_A)(1+b_B P_B)}$$

Biomolecular reaction

$$A \xrightarrow{I} P$$

Inhibitor

$$Rate = k_r P_A \theta_A = \frac{k_r P_A b_A P_A}{(1+b_A P_A + b_I P_I)}$$

dissociative adsorption

$$A_2(g) + 2S(s) \rightleftharpoons 2A{:}S(s)$$

$t=0$	P_{A_2}	2	0
at eqm	P_{A_2}	$2(1-\theta)$	2θ

r_{ads}

$$r_{ds} = k_d \theta^2$$

$$r_{ads} = k_a P_{A_2} (1-\theta)^2$$

$$\boxed{\theta = \frac{(b_{A_2} P_{A_2})^{1/2}}{1 + (b_{A_2} P_{A_2})^{1/2}}}$$

Similarly if it is getting in x fragment then

$$\boxed{\theta = \frac{(b_{A_2} P_{A_2})^{1/x}}{1 + (b_{A_2} P_{A_2})^{1/x}}}$$

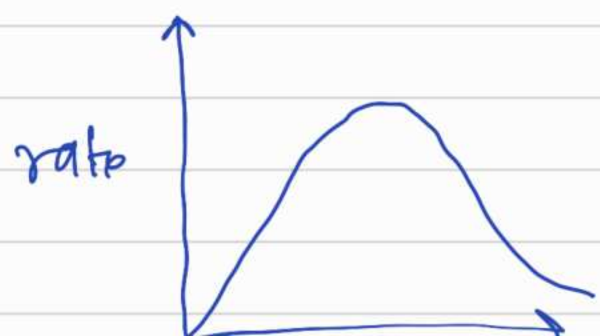
$$CO(g) \longrightarrow CO(ads)$$

$$O_2(g) \longrightarrow O_2(ads) \longrightarrow 2O(ads)$$

L-H mechanism

$$CO(ads) + O(ads) \longrightarrow CO_2(g)$$

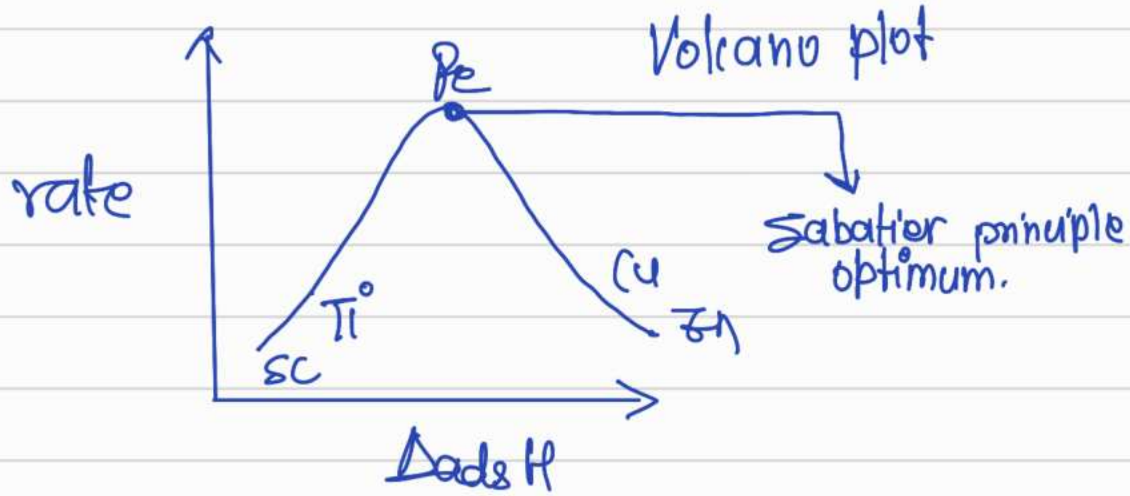
$$rate = k_r \theta_{CO} \theta_{O_2} = \frac{k_r b_{CO} P_{CO} (b_{O_2} P_{O_2})^{1/2}}{1 + b_{CO} P_{CO} + (b_{O_2} P_{O_2})^{1/2}}$$



P_{CO}

$N_2(g) \longrightarrow N_2(ads) \longrightarrow 2N(ads)$

$H_2(g) \longrightarrow H_2(ads) \longrightarrow 2H(ads)$



Sabatier principle

Physical interpretation of k_d

$A + S \longrightarrow AS$

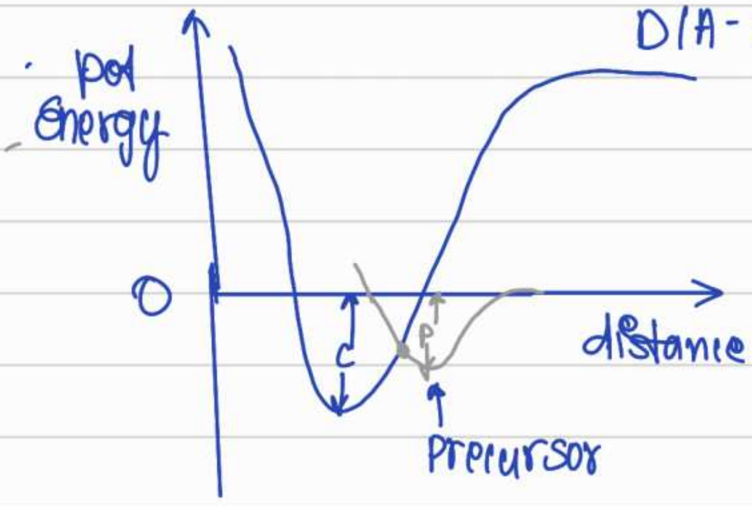
rate of adsorption or desorption.

$r = k_d \theta_A \Rightarrow k_d = A \cdot e^{-E/RT}$

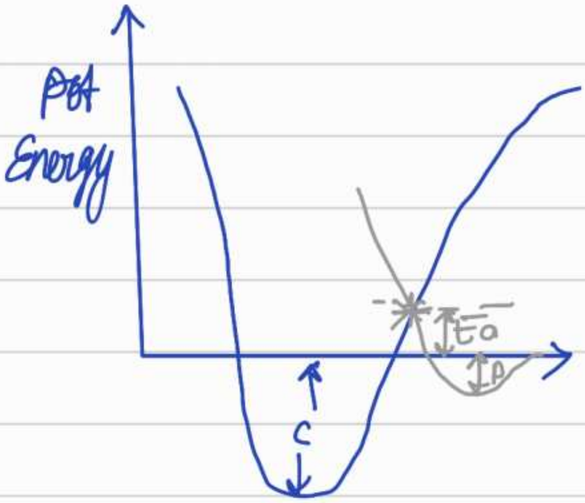
$\frac{1}{k_d} = \tau =$ residence time.

Temperature progress dissorption (TPD)

" " Rexn. (TPR)



Non- activated



Activated

Desorption ⇒ Always activated.

$$\text{Rate} = R = \frac{d\theta}{dt} = A'\theta^{x}$$

$k_d = \tau_0^{-1} e^{\frac{-E_{ads}}{RT}}$ → frequency factor.

Arrhenius-type equation.

$\tau = \tau_0 e^{E_{ads}/RT}$

└ Residence time.

Temperature programmed desorption (TPD)

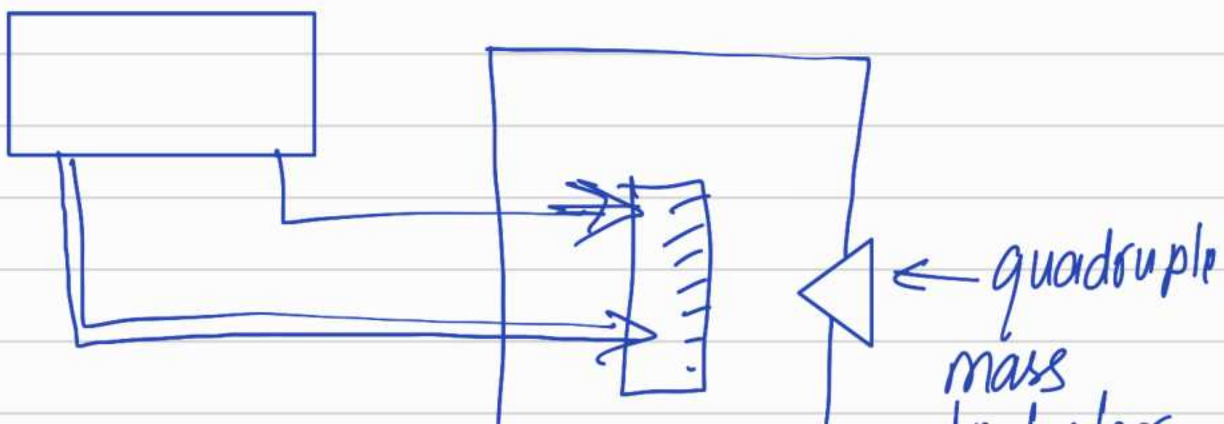
$T = T_0 + \alpha T$ ← Time

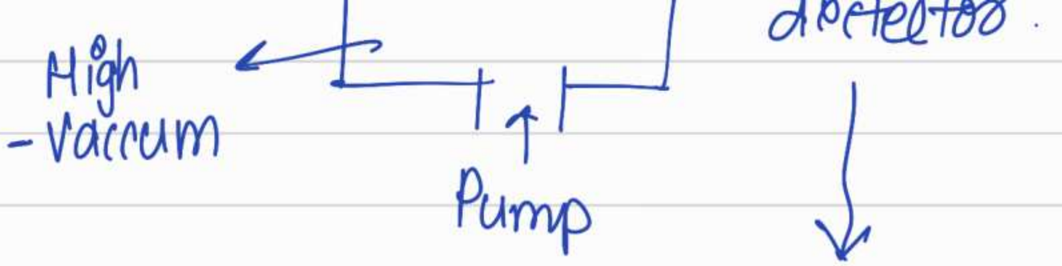
↑

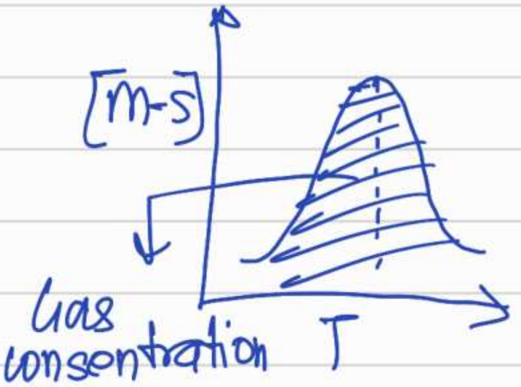
Heating rate

Liquid N_2 -- $-80°C$

Silican carbide







Rate of desorption

$$R = -\frac{d\theta}{dt} = k_d\,\theta^x$$

$$= \tau_0^{-1} e^{-E_a/RT}\,\theta^x$$

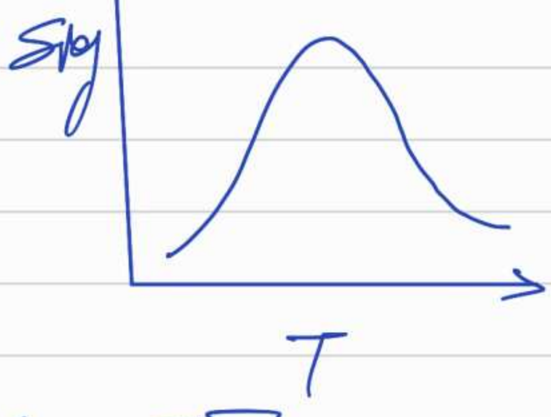
$$M + S \underset{k_{des}}{\overset{k_{ads}}{\rightleftharpoons}} M\text{-}S$$

$$\frac{d[M\text{-}S]}{dT} = -\frac{[M\text{-}S]}{\alpha}\left(\tau_0^{-1} e^{-E_a/RT}\right)$$

$$\frac{d[M\text{-}S]}{dT} = \frac{d[M\text{-}S]}{dt}\,\frac{dt}{dT} \quad \frac{1}{\alpha}$$

$$\frac{d[M\text{-}S]}{dT} = -\frac{\tau_0^{-1}}{\alpha}[M\text{-}S]\,e^{-E_a/RT}$$

$$\frac{d}{dT}\left(\frac{d[M\text{-}S]}{dT}\right) =$$



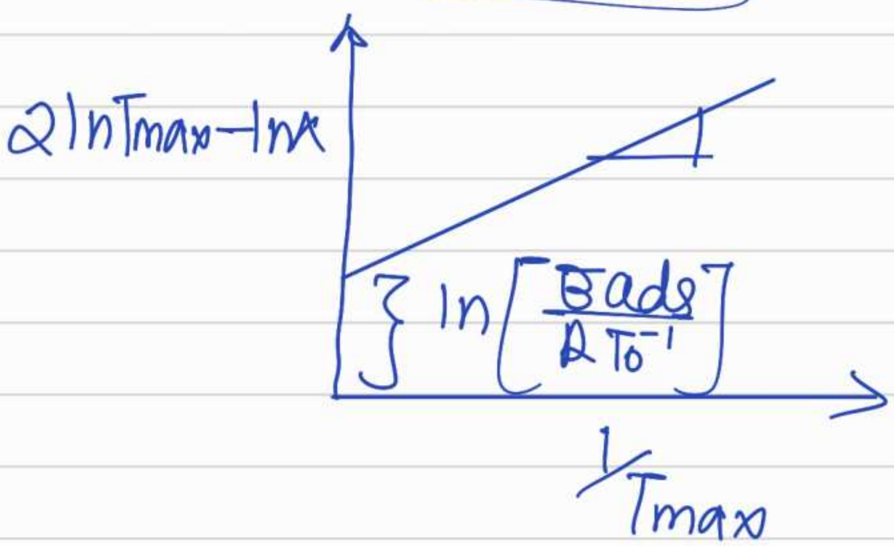
$$\neq \tau_0^{-1}[M\text{-}S]\,e^{-E_a/RT}\left(\ldots\right]^T$$

$$\frac{\tau_0}{\alpha}[M\text{-}S]\, e^{-E_a/RT}\left(\frac{E_a}{R}\left(\frac{-1}{T^2}\right)\right)$$

$$-\cancel{\frac{\tau_0^{-1}}{\alpha}}\,\frac{d[M\cdot S]}{dt}\,\cancel{e^{-E_a/RT}} = 0$$

$$\frac{-\tau_0}{\alpha}\left[\frac{[M\text{-}S]\,E_a}{RT^2} + \frac{d[M\text{-}S]}{dT}\right] = 0$$

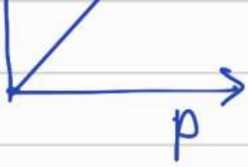
$$\boxed{2\ln T_{max} - \ln\alpha = \ln\left[\frac{E_{ads}}{R\,T_0^{-1}}\right] + \frac{E_{ads}}{R\,T_{max}}}$$



Langmuir adsorption isothermal.

$$\theta = \frac{bP}{1+bP}$$

$\theta = pb$ at low p



Chemisorption in practice

⇒ Heterogeneous adsorption sites Fractals, enthalpy

- Enthalpy and extent of adsorption are interdepended.

$$\boxed{\Theta(P,T) = \int_0^\infty \underbrace{\theta(P,T,E_{ads})}_{\text{Langmuir}} \underbrace{f(E_{ads})}_{f(E_{ads}) = \alpha e^{-E_{ads}/RT}} dE_{ads}}$$

on solving

Langmuir adsorption isotherm

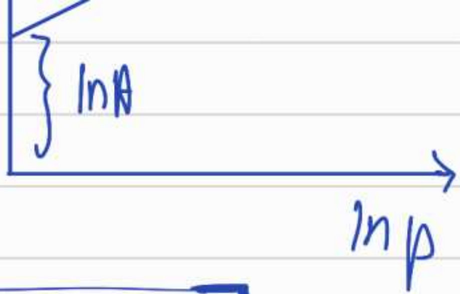
⇒ $\Theta = AP^c = AP^{1/n}$
freundlich equation.

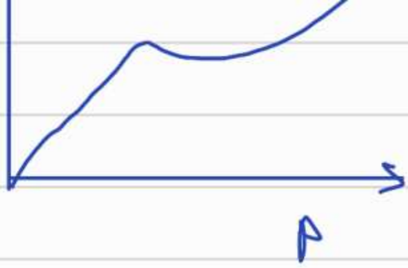
$$\Theta = \frac{V}{V_m} = AP^{1/n}$$

vol. of diff P

$$\ln V/V_m = \ln A + \frac{1}{n}\ln P$$

mono layer volume





$$\boxed{\theta = \frac{AP^{1/n}}{1 + AP^{1/n}}}$$

Temkin

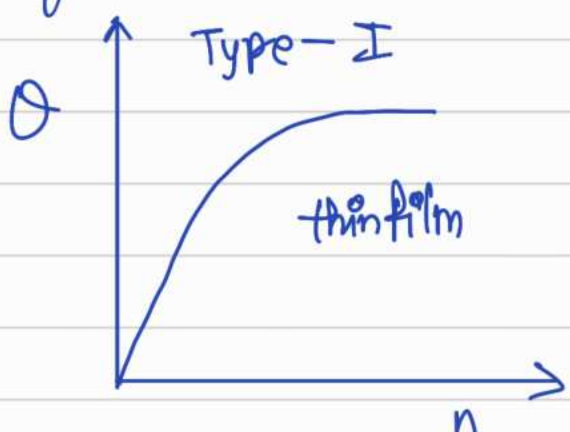
linear decrease of E_{ads}

$$F(E_{ads}) = E^{0}_{ads}(1 - \alpha\theta)$$

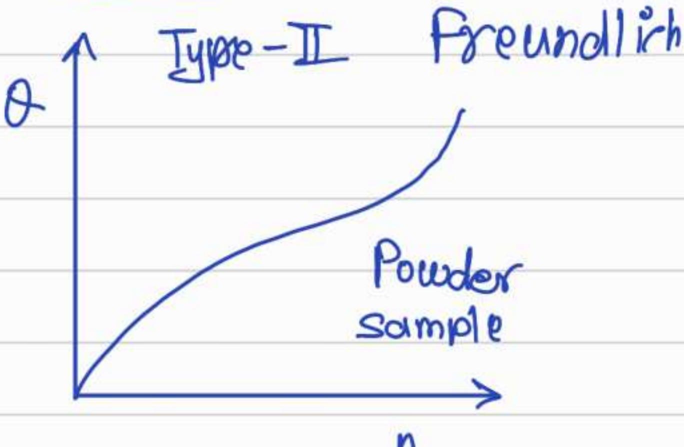
Linear decrease of E_{ads}

$$\boxed{\theta = \frac{RT}{E_{ads}\,\alpha} \ln P + \text{constant}}$$

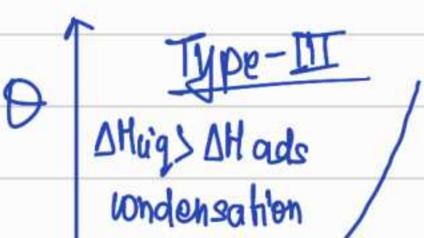
Type of adsorption isotherm.

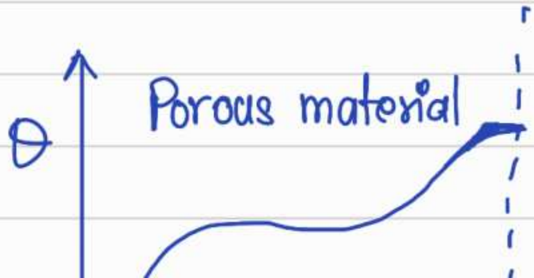


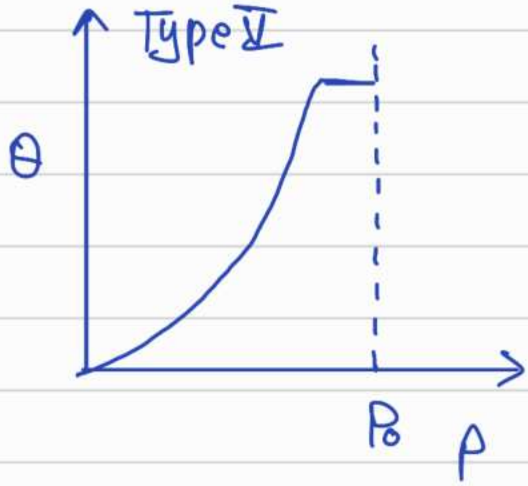
Chemisorption, monolayer



Physisorption, multilay







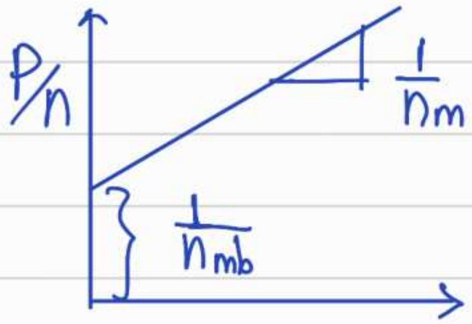
Termodynamics of adsorption

$$\theta = \frac{bP}{1+bP}$$

$$\theta = \frac{n}{n_m}$$ ← moles/g (for n)

← moles/g for monolayer (for n_m)

$$\frac{n}{n_m} = \frac{bP}{1+bP} \Rightarrow \boxed{\frac{P}{n} = \frac{1}{n_m b} + \frac{P}{n_m}}$$



$b = K_d$ = adsorption equilibrium constant.

$\Delta G = -RT \ln K_0$

$\Delta G = \Delta H - T\Delta S$

$$\ln K_d = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}$$

$$\Downarrow$$

$$\left[\frac{\partial}{\partial T}(\ln K_0)\right]_\theta = \frac{\Delta H}{RT^2} \Big\} \rightarrow \text{Vant Hoff eq}^n$$

Isosteric enthalpy of adsorption.

$$\theta = \frac{KP}{1+KP}$$ at fixed θ, $\ln P + \ln K = \text{const.}$

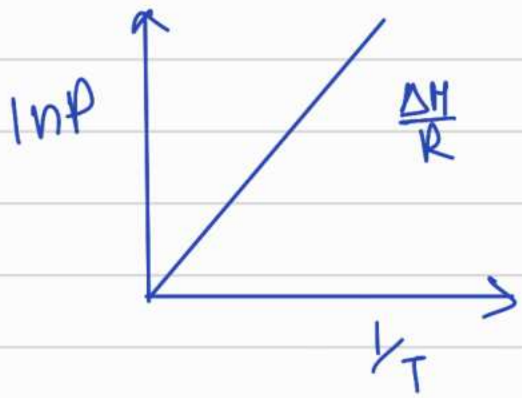
$$\Downarrow$$

$$KP = \frac{\theta}{1-\theta}$$

$$\ln P = \text{const} - \ln K$$

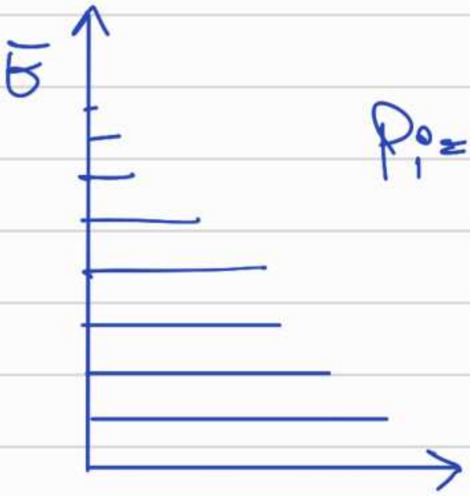
$$\left[\frac{\partial}{\partial T}(\ln P)\right]_\theta = \frac{-\Delta H}{RT^2}$$

$$\frac{\partial(1/T)}{\partial T} = \frac{-1}{T^2} \Rightarrow \frac{\partial \ln P}{\partial(1/T)} = \frac{\Delta H}{R}$$



Derivation of Langmuir Isothermal using statistical thermodynamics

Statistical thermodynamics approach



$$P_i = \frac{n_i}{N} = \frac{e^{-\varepsilon_i/k_bT}}{\sum_i e^{-\varepsilon_i/k_bT}}$$

$\varepsilon_n = \frac{n^2h^2}{8mL^2}$

$\big\downarrow$ for degeneracy

$$= \frac{e^{-\varepsilon_i/k_bT}}{\sum_i g_i e^{-\varepsilon_i/k_bT}}$$

$\frac{1}{kT} = \beta$

$$P_i = \frac{e^{-\beta\varepsilon_i}}{\sum_i g_i e^{-\beta\varepsilon_i}}$$

$$q = \underbrace{\sum_i g_i e^{-\beta\varepsilon_i}}_{\text{Partition function}}$$

For this

$$q^T = \sum_i g_i e^{-\beta\varepsilon_i}$$

$$= \sum_i g_i e^{-\left(\frac{n^2h^2}{8mL^2}\right)\beta}$$

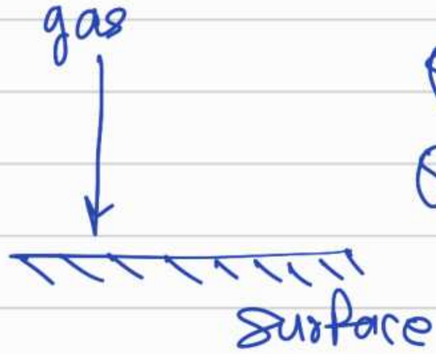
Reference $n=1$

$\varepsilon_n = (n^2-1)\varepsilon$

$$q^T = \int_1^\infty g_i e^{-(n^2-1)\beta\varepsilon}\,dn$$

$$\boxed{q_x^T = \left[\frac{2\pi m}{h^2\beta}\right]^{1/2} L}$$

For 1 dimension



$$Q^g = Q_T^g Q_{int}^g$$

$$Q^s = Q_T^s Q_{int}^s e^{-E_{ads}/RT}$$

$$Q_{ads}^g = Q_{int}^g Q_{ads}^g$$

$$S = k_b \ln W$$

$$E_{Total} = \varepsilon_x + \varepsilon_y + \varepsilon_z = q_x^T q_y^T q_z^T$$

$$q_{3D}^T = \left[\frac{2\pi m}{h^2\beta}\right]^{3/2} L^3$$

Helmohotlz energy

$$A^s = -RT\ln Q_{total}^s$$

$$= -RT\ln\left(Q_T^s Q_{int}^s e^{-E_{ads}/RT}\right)$$

$$A^{+} = -RT\ln(Q^{g}_{Total})$$
$$= -RT\ln(Q^{g}_{T}\, Q^{g}_{ads})$$

$\boxed{\text{vibration energy} = (n+\tfrac{1}{2})h\nu}$

$$\frac{\theta}{1-\theta} = \frac{Q^{s}}{Q^{g}} \Rightarrow \frac{\theta}{(1-\theta)} = \boxed{\frac{\left(\frac{h^2}{2\pi m kT}\right)^{3/2} Q^{s}_{T}\, Q^{s}_{in}\, e^{-\frac{\varepsilon_{ads}}{RT}}\, p}{RT\, Q^{g}_{int}}}$$

$$\frac{\theta}{1-\theta} = bp$$

$$\boxed{\theta = \frac{bp}{1+bp}}$$

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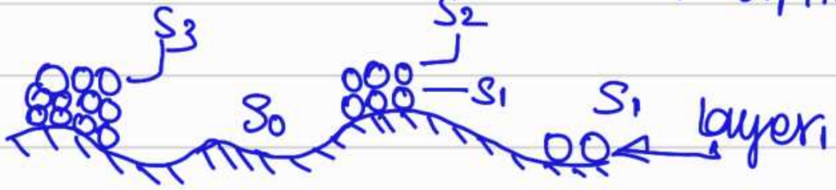
19 Oct

BET isotherm

⇒ Assumptions : Similar to Langmuir
- Homogeneity
- Adsorbate fixed on surface
- At equilibrium ⇒ Rate of adsorption = Rate of desorption.



- Adsorption may or may not be activated
- disorption is always activated.

Total site ⇒ $S = s_0 + s_1 + s_2 + \cdots + s_n = \sum_{i=0}^{n} s_i$

$i=0$

Bare Surface

Rate of ads = Rate of desorption

$$k_a P S_0 = k_{d_1} S_1 \Rightarrow S_1 = \left[\frac{k_a}{k_{d_1}} P\right] S_0$$

Assuming rate constant is same

$$k_a P S_1 = k_{d_2} S_2 \Rightarrow S_2 = \left[\frac{k_a}{k_{d_2}} P\right] S_1$$

$$\left(k_{d_2} = k_{d_3} \cdots \right)$$

$$S_2 = \left[\frac{k_a^2 P^2}{k_{d_1} k_{d_2}}\right] S_0$$

$$S_3 = \left[\frac{k_a^2 P^2}{k_{d_2}^2}\right] S_1$$

$$\boxed{S_i = \left[\frac{k_a^{i-1} P^{i-1}}{k_{d_2}^{i-1}}\right] S_1}$$

$$\boxed{S_i = \left[\frac{k_a^{i} P^{i}}{k_{d_2}^{i-1} k_{d_1}}\right] S_0}$$

$$k_d = \nu e^{-E_{ads}/k_B T}$$

$$k_{d_1} = \nu e^{-E_{ad}/k_B T}, \quad k_{d_2} = \nu e^{-E_v/k_B T}$$

$$S_i = \left[\frac{k_a P}{\nu e^{-E_v/k_B T}}\right]^i e^{(E_{ad} - E_v)/k_B T} S_0$$

$$\underbrace{\qquad}_{x} \qquad \underbrace{\qquad}_{C} \qquad \underbrace{\qquad}_{S_0}$$

$$S_i = x^i C S_0$$

$$\sum_{i=1}^{\infty} S_i = C S_0 \sum_{i=1}^{\infty} x^i = \frac{C S_0 x}{1-x}$$

$$S = S_0 + S_1 + S_2 + \cdots + S_n$$

$$S = S_0 + \sum_{i=1}^{n} x^i C S_0$$

Let V = Total volume of adsobed gas on surface.

V_m = Volume for mono layer formation

$$V = \sum_{i=1}^{n} V_i \propto \sum_{i=1}^{n} i S_i = \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}$$

$$V_m \propto S \Rightarrow V_m = S_0 + \sum_{i=1} x^i C S_0$$

$$\frac{V}{V_m} = \theta = \frac{Cx}{(1-x)[1+(C-1)x]}$$

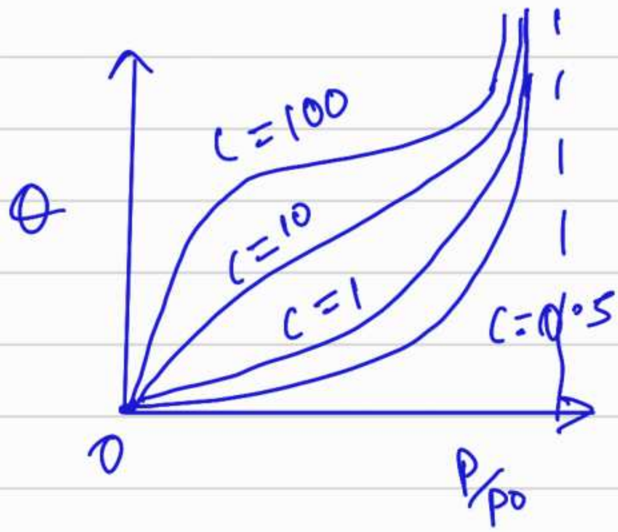
$$C = e^{(E_{ads} - E_V)/k_B T} \, ;$$

$$V \to \infty$$

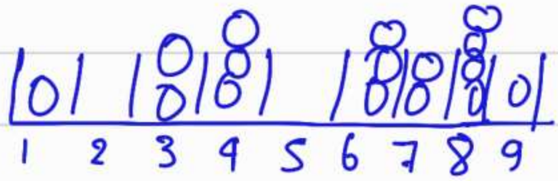
$$x = 1 \Rightarrow x = \frac{K_a P}{\ldots} = \frac{K_{ai} P}{\ldots}$$

$\nu e^{-\epsilon v/k_BT}$ $\quad$ $K_d, i=2$

If $p \to p_0$, $x = p/p_0$



BET



$\theta_0 = \frac{2}{9}$

$\theta_1 = 2/9$

Similarly $\theta_2 \ldots$

$$\theta_i = \frac{k_{ai}}{k_{di}} P \theta_{i-1}$$

$$= \frac{k_{a\,i-1}}{k_{d\,i-1}} P \frac{k_a}{k_d} P \theta_0$$

$$x = P\frac{k_{ai}}{k_{di}} \Rightarrow x^i = P^i \frac{k_{ai}}{k_{di}}$$

$$\theta_i = x^i c\, \theta_0$$

$$\left(\theta = \frac{\text{number of molecules of sites}}{\text{Total number of sites.}} \right)$$

$$x = p\frac{k_{ai}}{k_{di}}$$

$$\theta = \frac{n_a}{n}$$

At equilibrium

At equilibrium,

pressure $(P) = P_0$

$\downarrow$

eqm vapour pressure.

$n_a = \sum_{i=1}^{\infty} i\theta_i N$

$= \sum_{i=1}^{\infty} i x^i$

$= \simeq \frac{x}{(1-x)^2}$

$K_{ai} P_0 N = K_{di} N$

$P_0 = \frac{K_{di}}{K_{ai}}$

$\left\{ x = \frac{P}{P_0} < 1 \right\}$

$$C = \frac{P}{x}\frac{K_a}{K_{ad}} = \frac{P}{x}\frac{e^{-\bar{E}_{ads}/RT}}{e^{-\bar{E}_v/RT}} = e^{(\bar{E}_{ads}-\bar{E}_v)/RT}$$

$$= e^{(\Delta_{ads}H^0 - \Delta_{vap}H^0)/RT}$$

Specific Surface Area.

$$A_{sp} = \frac{\text{Total Area of Sample}}{\text{Total mass.}}$$

$$\theta = \frac{n^s N_A \sigma^0}{W \cdot A_{sp}} \Rightarrow \theta = 1 \text{ at saturation.}$$

Foot print $= \sigma^0 =$ Area of molecule.

$$A_{sp} = \frac{n^s N_A \sigma^0}{}$$

w

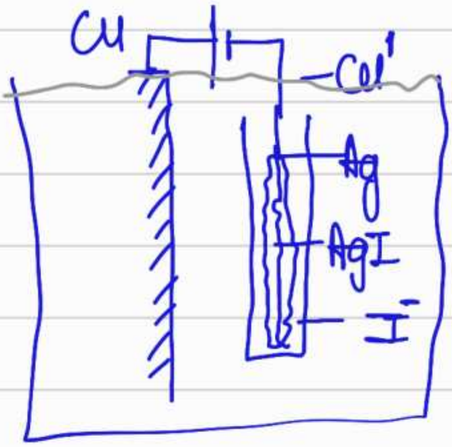
Limitation = $\left(\dfrac{P}{P_0} = 0.05 \text{ to } 0.3\right)$

Electrified interfaces

Book ⇒ Pallab Ghosh
colloid and
interfaces

AgI → negative charge tendency
I = chemically adsorption.

1> Adsorption of ions
2> dissociation of surface functionality.



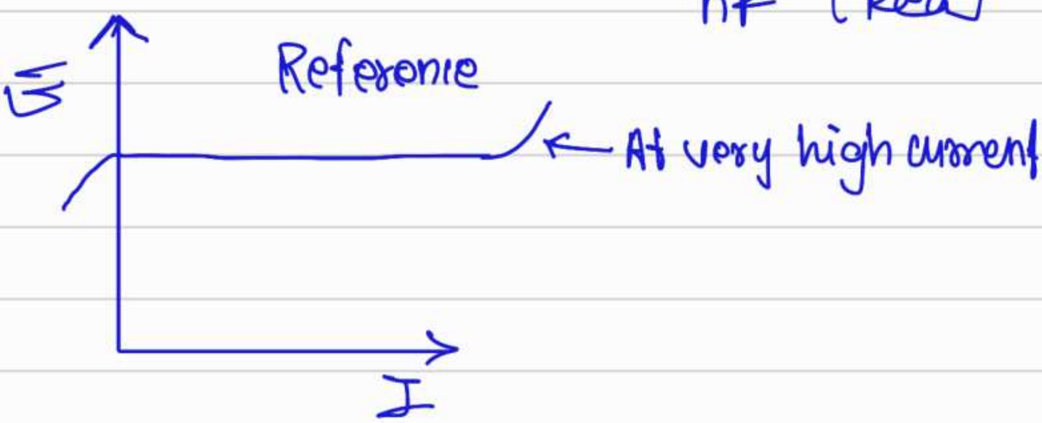
Ag / AgI , I^-(aq)

Reversible electrode

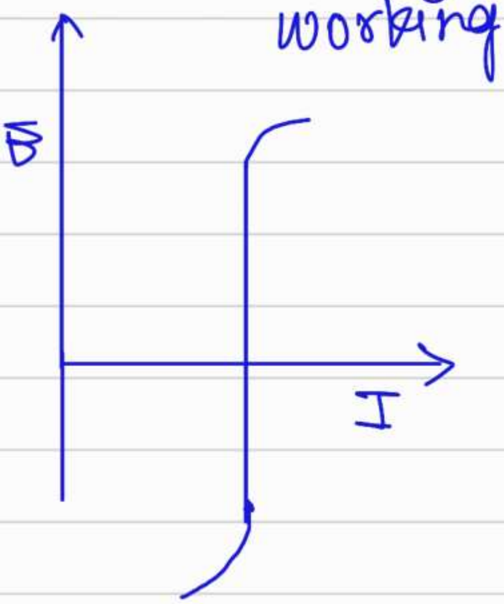
Ideally non-polarizable electrode

$$Ox + e^- \rightleftharpoons Red$$

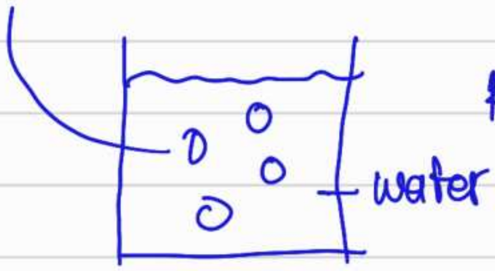
Nernst eq^n = $E = E^0 + \dfrac{RT}{nF} \ln\left[\dfrac{Ox}{Red}\right]$



Ideally polarizable electrode



AgI particle



$K_{sp} = 7.5\times10^{-17}$ at 25°C

$[Ag] = [I] = 8.5\times10^{-9}$

→ Specifically adsorbed or preferentially adsorbed.

point of zero charge., $\overset{3.0\times10^{-9}}{C_{PZC}} \rightarrow$ conc. → PZC = 0V
potential of zero charge.

Taking this reference

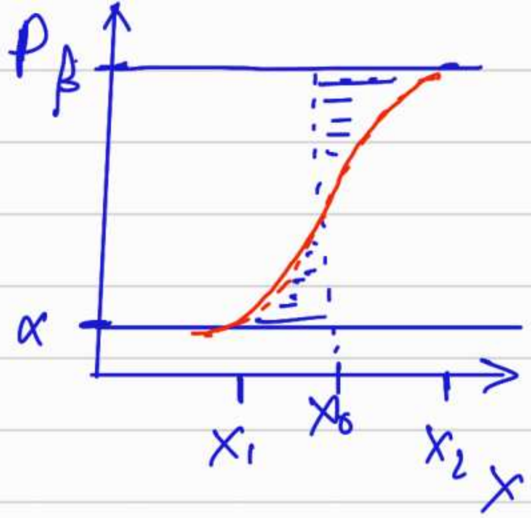
$$\boxed{\psi_0 = 2.303\,\frac{RT}{nF}\log\left[\frac{C}{C_{PZC}}\right]}$$

if $[Ag^+] = 8.5\times10^{-9}$

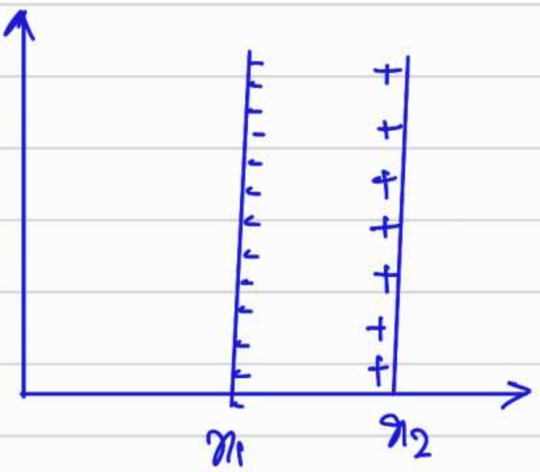
$\psi_0 = -150$ mV.

C_{PZC} → upper kuch aur
→ niche kuch aur.

Gibbs adsorption isotherm

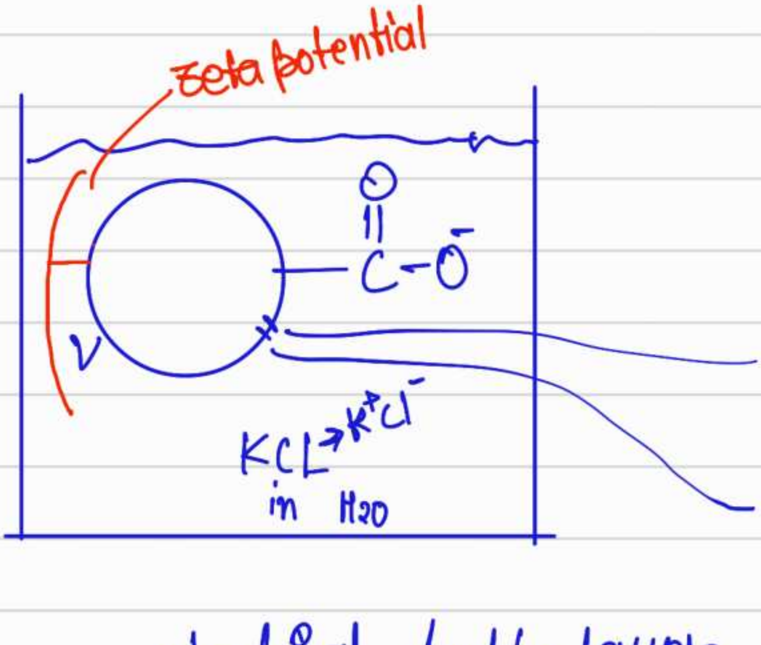


Helmholtz Model

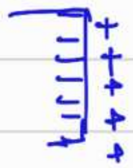


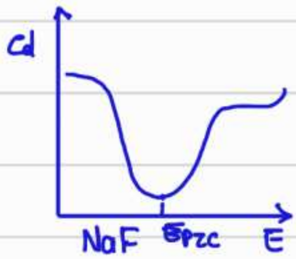
colloidal
↳ nano particle

quantum dot



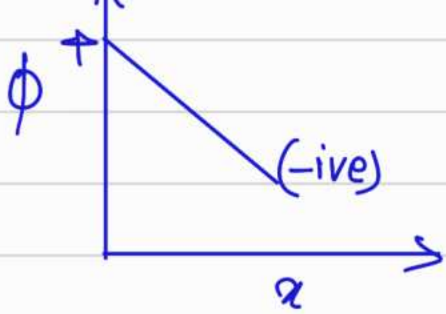
$\sigma_1^m = -\sigma^s$ ↪ should be equal & opposit





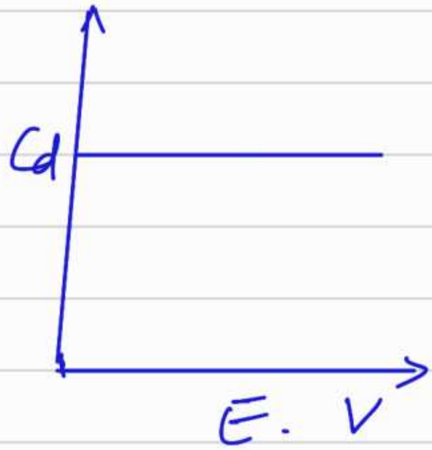
electrical double layer.

$$\sigma^m = \frac{q}{A} = \frac{\varepsilon \varepsilon_0 V}{d}$$

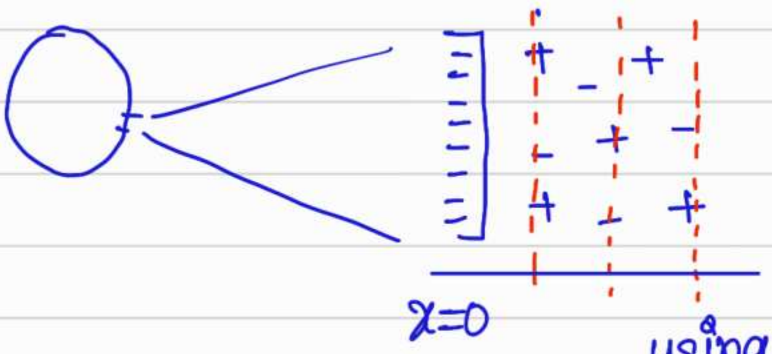


$$\text{Capacitance} = \frac{\text{Charge}}{\text{Potential}}$$

$$C_d = \frac{d\sigma^m}{dV} = \frac{\varepsilon \varepsilon_0}{d}$$



Gouy - Chapmann (Diffuse double layer).



Total no. of ions $= n^0$

Ions in layer $= n_i$

using Boltzman distribution,

$$n^0 = n_i \exp\left(\frac{-z_i e \phi}{kT}\right)$$

z_i = charge on ion

e = electron charge

ϕ = potential

poisson's eq^n

$$\frac{d^2\phi}{dx^2} = \frac{-\rho}{\varepsilon \varepsilon_0}$$

$$\rho = \sum n_i z_i e = \sum n_i z_i e \exp\left(\frac{-z_i e \phi}{kT}\right)$$

charge

density

$$\boxed{\frac{d^2\phi}{dx^2} = \frac{-e}{\varepsilon\varepsilon_0}\sum_i n_i z_i \exp\left(-\frac{z_i e\phi}{kT}\right)}$$

(i) → If z:z electrolyte symmetrical.

$$\frac{d^2\phi}{dx^2} = \frac{-e}{\varepsilon\varepsilon_0}\left[n_i^{\circ} z \exp\left(\frac{-ze\phi}{kT}\right) - n_i^{\circ} z \exp\left(\frac{ze\phi}{kT}\right)\right]$$

$$= \frac{-e n_i^{\circ} z}{\varepsilon\varepsilon_0}\left[\exp\left(\frac{-ze\phi}{kT}\right) - \exp\left(\frac{ze\phi}{kT}\right)\right]$$

(ii) For small ϕ, $e^{\pm x} = 1 \pm x$

$K^2 = \frac{1}{L^2}$ = inverse of Debye. length

$$= \frac{2e^2 n_i^{\circ} z^2 \phi}{\varepsilon\varepsilon_0 kT} = K^2\phi$$

जहां तक charge का influence रहता है!

$$K^2 = \frac{2e^2 n_i z^2}{\varepsilon\varepsilon_0 kT}$$

<u>for Assymetrical electrolytes.</u>

$$\frac{d^2\phi}{dx^2} = \frac{-\rho}{\varepsilon\varepsilon_0} = \frac{-e}{\varepsilon\varepsilon_0}\sum n_i z_i \exp\left(\frac{-z_i e\phi}{kT}\right)$$

$$= \frac{-e}{\varepsilon\varepsilon_0} \underbrace{\sum n_i^0 z_i}\left(1 - \frac{z_i e \phi}{kT}\right)$$

$$= \boxed{\frac{e^2}{\varepsilon\varepsilon_0 kT} \sum n_i^0 z_i^2} \phi \;\rightarrow\; = \frac{e^2 N_A}{\varepsilon\varepsilon_0 kT} \sum_i c_i z_i^2 \phi$$

$$\frac{d^2\phi}{dx^2} = k^2 \phi$$

$$\text{no. of moles} = \frac{\text{Total no. of ions}}{N_A}$$

$$C = \frac{n_i^0}{N_A} \Rightarrow n_i^0 = c_i N_A$$

at 25°C; for symmetrical

$k(m^{-1})$ $3.29 \times 10^9 z_i \, c_i^{1/2}$ c = moles/Lit.

$L_D = \frac{1}{k}$, $3.04 \times 10^{-10} |z|^{-1}$

for assymmetrical

$$k(m^{-1}) = 2.32 \times 10^9 \sum |z_i| c_i^{1/2}$$

$$L_D = 4.30 \times 10^{-10} \sum_i |z_i|^{-1} c^{-1/2}$$

$$\frac{d^2\phi}{dx^2} = k^2 \phi$$

$\overline{dx^2}$

$$\phi = Ae^{kx} + Be^{-kx}$$

Applying boundary condn,

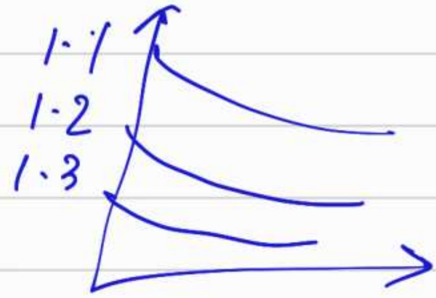
$x = 0$ $x = \infty$

$\phi = \phi_0$ $\phi \to \infty$

NaCl
$MgCl_2$
$AlCl_3$

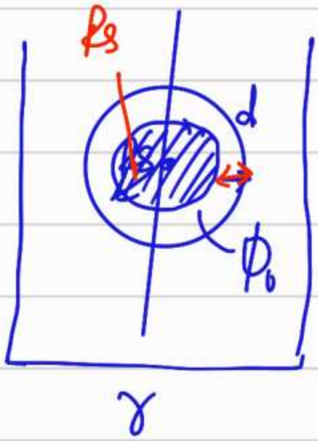
$$\phi = \phi_0 e^{-kx}$$

$$\frac{\phi}{\phi_0} = e^{-kx}$$



In spherical polar coordinate

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)\phi = k^2\phi$$



$$\phi = \frac{C_1}{r}e^{kr} + \frac{C_2}{r}e^{-kr}$$

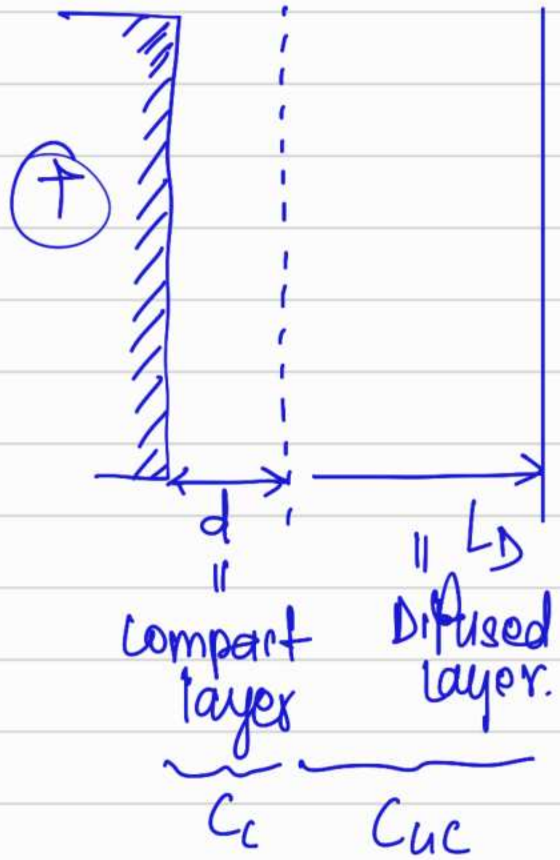
B.C, If $r = R_s + d$, $\phi = \phi_d$

$$C_1 = 0, \quad C_2 = \phi_d (R_s + d) e^{k(R_s + d)}$$

$$\phi = \frac{\phi_d (R_s + d) e^{k(R_s + d)}}{r} e^{-kr}$$

$$\frac{\phi}{\phi_d} = \frac{(R_s+d)}{r} e^{-k(r-R_s-d)}.$$

Stern model

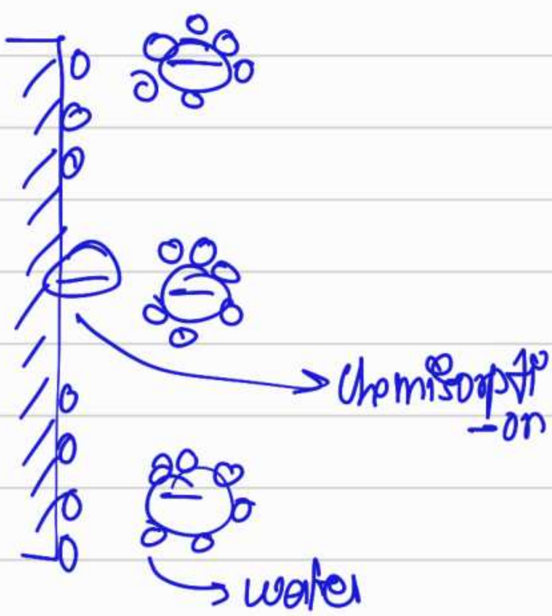


Bulk solution

$$\frac{1}{C_s} = \frac{1}{C_c} + \frac{1}{C_{uc}}$$

Grahame

BDM



First layer | layer → OHP

↓

IHP

Inner Helmholtz Plane

$$\frac{\partial^2 \phi}{\partial x^2} = -\frac{\rho}{\varepsilon\varepsilon_0} = \frac{-e}{\varepsilon\varepsilon_0}\sum_i n_i^0 z_i \exp\left(\frac{-z_i e\phi}{kT}\right)$$

$$\frac{d}{d\phi}\left(\frac{d\phi}{dx}\right)^2 = \frac{-2e}{\varepsilon\varepsilon_0}\sum n_i^0 z_i \exp\left(-\frac{z_i e\phi}{kT}\right)$$

Integrating

$$\left(\frac{d\phi}{dx}\right)^2 = \frac{2e}{\varepsilon\varepsilon_0}\sum n_i^0 z_i \frac{kT}{z_i e} \exp\left(\frac{-z_i e\phi}{kT}\right) + I$$

$$= \frac{2kT}{\varepsilon\varepsilon_0}\sum_i n_i^0 \exp\left(\frac{-z_i e\phi}{kT}\right) + \underbrace{I}_{\text{constant}}$$

<u>After applying Boundary condⁿ</u>

For z:z symmetrical electrolyte.

$$\left(\frac{d\phi}{dx}\right) = \left[\frac{2kTn^0}{\varepsilon\varepsilon_0}\right]^{1/2}\left[\exp\left(\frac{-ze\phi}{2kT}\right) - \exp\left(\frac{ze\phi}{2kT}\right)\right] \quad \text{—(i)}$$

$$\sigma^m = -\sigma^s$$

$$\sigma^m = -\int_0^\infty \rho\, dx$$

$$\sigma^d = -\int_d^\infty \rho\, dx = \varepsilon\varepsilon_0 \int_d^\infty \left(\frac{d^2\phi}{dx^2}\right) dx$$

$$= \varepsilon\varepsilon_0 \left[\frac{d\phi}{dx}\right]_d^\infty$$

$$= \varepsilon\varepsilon_0 \left[\underbrace{\left(\frac{d\phi}{dx}\right)_\infty}_{0} - \left(\frac{d\phi}{dx}\right)_d\right]$$

$$\sigma^d = -\varepsilon\varepsilon_0 \frac{d\phi_d}{dx}$$

from (i)

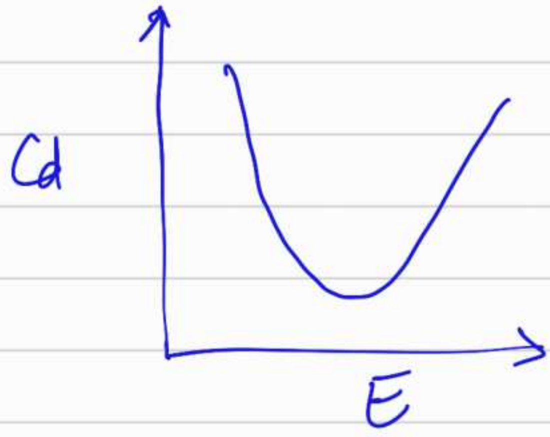
$$\sigma^d = -\varepsilon\varepsilon_0 \left(\frac{2kTn^0}{\varepsilon\varepsilon_0}\right)^{1/2} \left[\exp\left(\frac{-ze\phi}{2kT}\right) - \exp\left(\frac{ze\phi}{2kT}\right)\right]$$

$$= \left[2kT\varepsilon\varepsilon_0 n^0\right]^{1/2} \left[\exp\left(\frac{ze\phi_d}{2kT}\right) - \exp\left(\frac{-ze\phi_d}{2kT}\right)\right]$$

$$\sigma^d = \left[8kT\varepsilon\varepsilon_0 n^0\right]^{1/2} \sinh\left(\frac{ze\phi}{2kT}\right)$$

$$C_d = \frac{d\sigma^d}{\ldots} = (\text{Some constant}) \cosh\left(\frac{ze\phi}{2kT}\right)$$

$d\phi$ (2F)

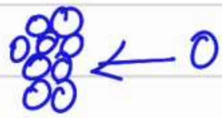


4 Nov

Colloidal Stability

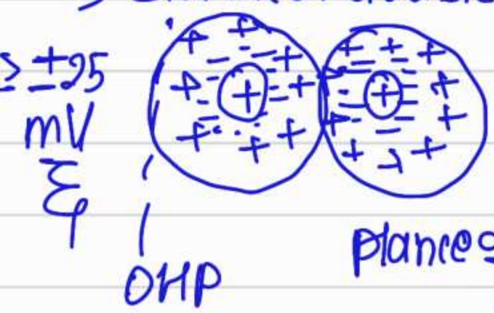
Faraday's Au. colloids

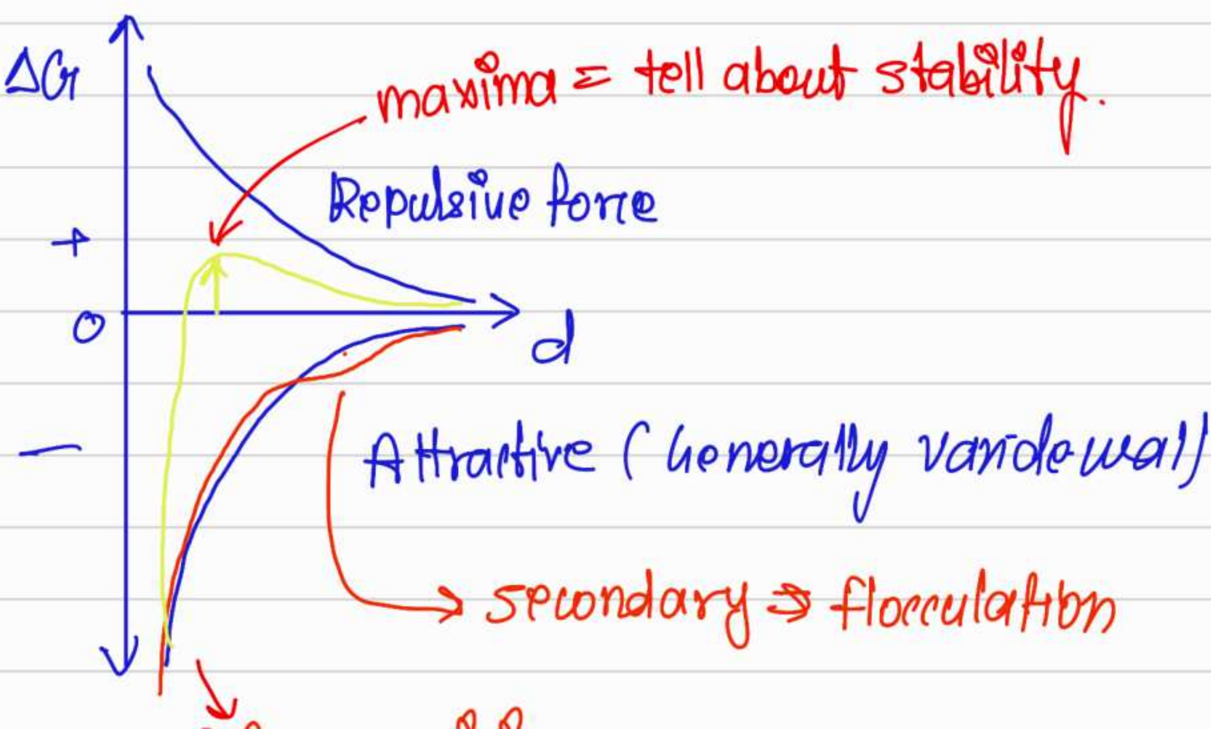
$\Rightarrow$ Way to stabilize



$\Downarrow$

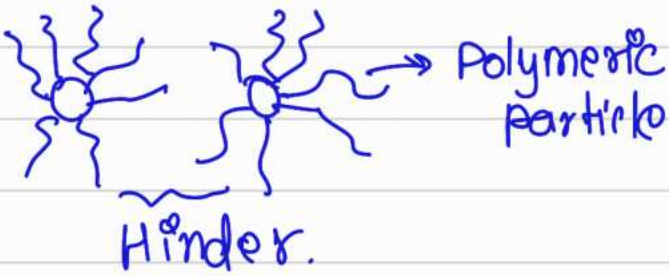
i) Electrical double layer





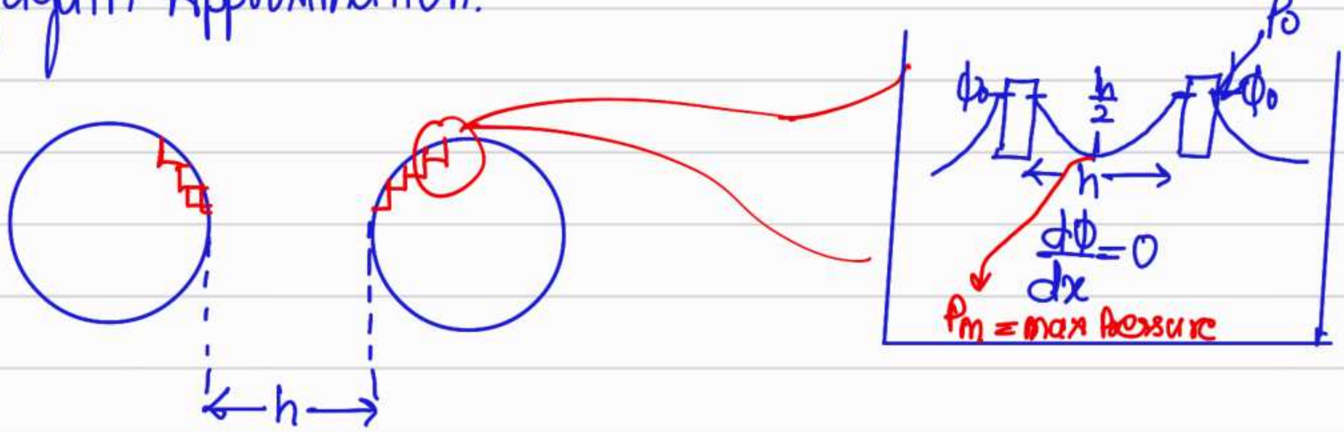
Primary minima
agglomeration

2) Steric Stabilization.

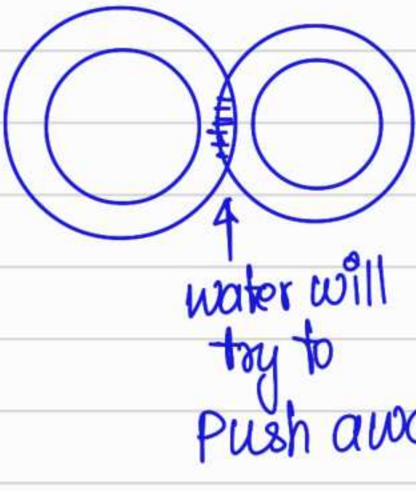


DLVO (Derjaguin-Landau-Verway-Overbeek).

Derjaguin Approximation.



Two forces



$$\phi = \frac{4kT}{ze}\tanh\left(\frac{ze\phi_0}{4kT}\right)\exp[-k(x-d)]$$

↓ derive this.

$$F_x = -\frac{dP}{dx}$$

$$F_{ele} = -\frac{\rho\, d\phi}{dx}$$

$$\rho = \sum n_i z_i e$$
$$= \sum n_i^0 z_i e \cdot \exp\left(-z_i e\phi \ldots\right)$$

$$= -\sum_i n_i^0 z_i e \exp\left(\frac{-z_i e \phi}{kT}\right) d\phi$$

for symmetrical

$$\rho = n^0 e z \left[\exp\left(\frac{-ze\phi}{kT}\right) - \exp\left(\frac{ze\phi}{kT}\right)\right]$$

$$\rho = -2 n^0 e z \sinh\left(\frac{ze\phi}{kT}\right)$$

$$= 2 n^0 e z \sinh\left(\frac{ze\phi}{kT}\right) d\phi$$

at equilibrium,

$$dP = -\rho\, d\phi$$

$$\int_{P_0}^{P_m} dP = 2 n^0 e z \int_0^{\phi_{h/2}} \sinh\left(\frac{ze\phi}{kT}\right) d\phi$$

$$P_{h/2} - P_0 = 2kTn^0\left[\cosh\left(\frac{ze\phi_{h/2}}{kT}\right) - 1\right]$$

Apply taylor series and take 2 terms for small $\phi_{h/2}$ potential.

$$\Pi_{EDL} = kTn^0\left(\frac{ze\phi_{h/2}}{kT}\right)^2 \text{ for small } \phi$$

$$\Pi_{EDL} = 64 kTn^0 \tanh^2\left(\frac{ze\phi_{h/2}}{4kT}\right) \cdot \exp(-\kappa h) \text{ for } x = \frac{h}{2}$$

Electrostatic force

6 Nov

$$\Phi_{net} = \Phi_{vDW} + \underbrace{\phi_{EDL}}_{=2\Phi}$$

$$d\Phi = -F_E\, dr \Rightarrow \text{integrate this after putting}$$

$$F_E = \Pi$$

electrostatic

$$\Phi_E = 64 kTn^0 \tanh^2\left(\frac{ze\phi}{\ldots}\right) \exp(-\kappa h)\, \kappa^{-1}$$

$$\Phi = 64\pi n \tanh\left(\frac{ze\psi_0}{4kT}\right) \exp(-\kappa h)\ \kappa$$

Electrophorisis

Electrical force

Viscous force $= F_v = 6\pi\eta R v$

At steady state, $QE = 6\pi\eta R v \Rightarrow \frac{u}{E} = \frac{Q}{6\pi\eta R}$

$\Rightarrow$ Electrophoretic mobility.

$$\phi = \frac{c_1}{r}\exp(-\kappa r) + \frac{c_2}{r}\exp(\kappa r)$$

at $r \to \infty$, $\phi = 0 \Rightarrow c_2 = 0$

$\phi = \frac{c_1}{r}\exp(-\kappa r) \Rightarrow$ at κ^{-1} - very large

$c_1 = \phi r$

$\phi = \frac{Q}{4\pi\varepsilon\varepsilon_0 r}$

$\hookrightarrow$ By gauss law

$Q = 4\pi\varepsilon\varepsilon_0 \phi r$

$\Downarrow$

$\frac{u}{E} = 4\pi\varepsilon\varepsilon_0 R$

$E \qquad \overline{6\pi\eta R}$

$$= \frac{2}{3}\frac{\varepsilon\varepsilon_0\phi}{\eta}$$

$\phi = \xi$ = zeta potential

$$\frac{u}{E} = \frac{2}{3}\frac{\varepsilon\varepsilon_0\xi}{\eta}$$

Ideal Gas

$PV = nRT$

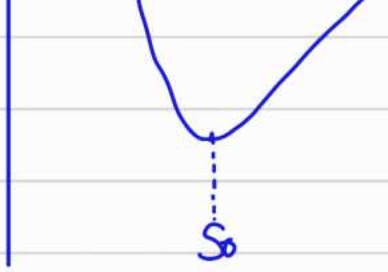
Real gas

$$\left(P + \frac{an^2}{V^2}\right)\left(V - nb\right) = nRT$$

Mie $\Rightarrow$ $\phi = -\frac{A}{s^p} + \frac{B}{s^q}$

London $\Rightarrow$ $\phi = -\frac{A}{s^{p}} + \frac{B}{s^{12}}$ Lennard-Jones potential.

L-J potential or 6-12 potential



$$A = 1.02 \times 10^{-77}\ \text{J m}^6$$
$$B = 1.57 \times 10^{-134}\ \text{J m}^{12}$$

Viva
11, 13, 14

$$\frac{d\phi}{ds} = 0$$

$$+6AS^{-7} - 12BS^{-13} = 0$$

$$S^{-7}\left(6A - 12BS^{-6}\right) = 0$$

$$\frac{6A}{12B} = S^{-6} \Rightarrow S_0 = \left(\frac{2B}{A}\right)^{\frac{1}{6}}$$

$$S_0 = 3.8 \times 10^{-10}\ \text{m}$$

$$\boxed{\phi_m = -1.65 \times 10^{-21}\ \text{J}}$$

7 nov

$$\phi_{VDW} = \frac{-A_{vdw}}{S^6} = -\frac{(\overbrace{A_L}^{\text{Dominating}} + A_K + A_D)}{S^6}$$

A_H = Hamaker constant

for planer surface

$$\Phi_{net} = \Phi_{EDL} + \Phi_{VDW} = 64kTn^0\kappa^{-1}\tanh^2\left(\frac{ze\phi_0}{4kT}\right)\exp(-\kappa h) - \frac{A_H}{12\pi h^2}$$

A_H = Hamaker constant

$$VDW = \frac{A_L + A_K + A_D}{S^6}$$

A_L = London dispersion force const. induced dipole - induced dipole.

A_K = keesom - " - Force two permanent dipole.

A_D = Debye - " - (Permanent dipole - induced dipole).

$A_H = \pi^2 n_1 n_2 A_L$

for two different types of molecule.

$$A_L = \frac{9ab}{4\pi^2 N_A^3}$$ → vander waal parameter in real gas eq^n.

eg. For CH_4, $a = 0.228\ m^6\ Pa\ mol^{-2}$

$b = 4.3 \times 10^{-5} m^3 mol^{-1}$

$A_L = 1.02 \times 10^{-77}\ J.\ m^3$. $1\ Pa = 1\ J m^6$.

$$A_L = \left[\frac{3\alpha_1 \alpha_2 h \mu_1 \mu_2}{2(\mu_1 + \mu_2)}\right]\left[\frac{1}{4\pi\varepsilon_0}\right]^2$$

α = polarisibility

If $\alpha_1 = \alpha_2$ (same molecule interaction)

$$A_L = \left[\frac{3\alpha^2 h\nu}{\ldots}\right]\left[\frac{1}{4\pi\varepsilon_0}\right]^2$$

Eg → Neon gas

$$h\nu = 3.46 \times 10^{-18} J, \quad \frac{\alpha}{4\pi\varepsilon_0} = 3.9 \times 10^{-31} m^3$$

$$A_L = 3.95 \times 10^{-79} J m^3$$

$$A_H = \pi^2 n_1 n_2 A_L^{1,2}$$

① ② in vacuum

$$A_H^{1,3,2} = A_H^{1,2} + A_H^{3,3} - A_H^{1,3} - A_H^{3,2}$$

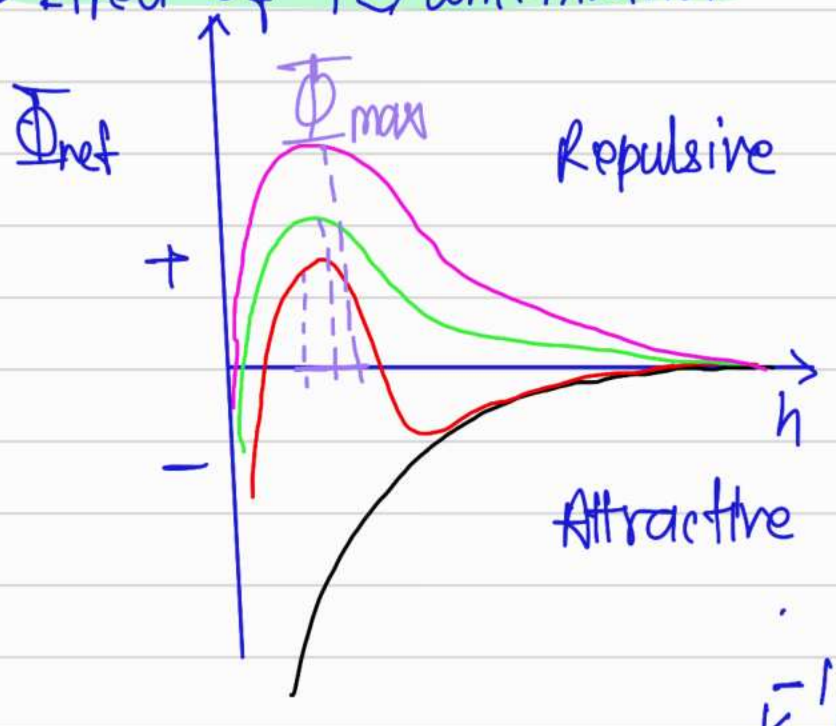
medium

① | ③ | ②

molecule of intrest

$$\Phi_{net} = \Phi_{EDL} + \Phi_{VDW} = \underbrace{64 k T n^0 \kappa^{-1} \tanh^2\left(\frac{z e \Phi_0}{4kT}\right)\exp(-\kappa h)}_{\text{Repulsive force}} \underbrace{- \frac{A_H}{12\pi h^2}}_{\text{Attractive}}$$

1) Effect of κ^{-1}/concentration

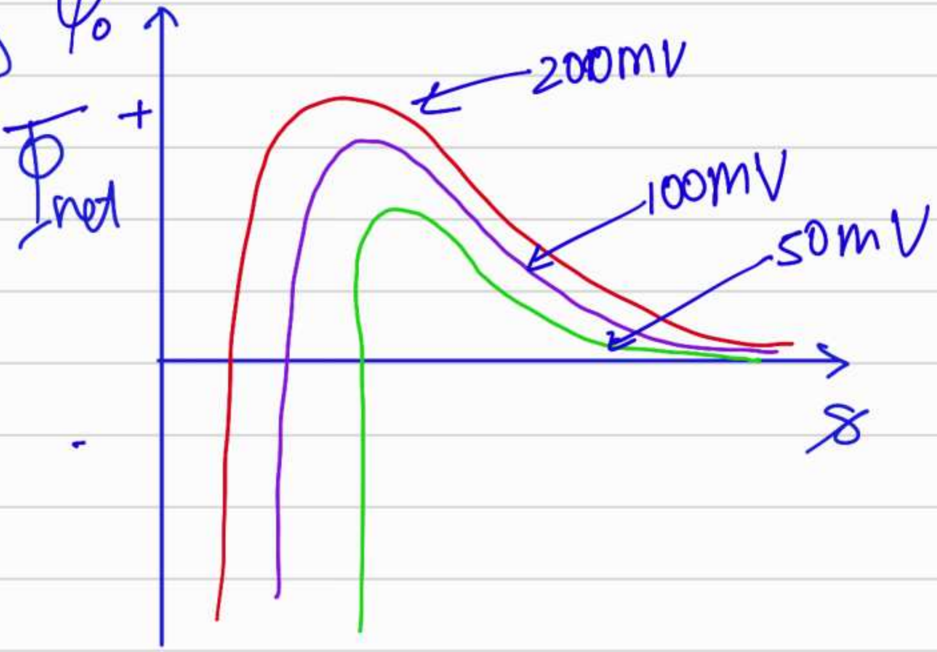


$\kappa^{-1}(10^{-6} cm) = 0.1 \Leftarrow 1$

$1.0 \Leftarrow 2$

3: 3 ← 3
10 ← 4

2) $\phi_0 \rightarrow$ with



3)